Name： **StudentID：**

Part A ER (Entity-Relationship) Diagram Drawing

Please draw the E-R Diagram. The descriptions below are sourced from the textbook.

A university plans to develop a database to manage its departments, instructors, students, courses, and course sections. The database will record the relationships among these entities, student enrollments, and the grades students receive.

Each department (DEPT) has a unique department name (DName) and a unique department code (DCode). No two departments may share either value. The database records the instructors, students, and courses associated with each department.

Each instructor (INSTRUCTOR) has a unique identifier (Id) and a phone number (IPhone). Every instructor must work for exactly one department. A department may employ zero or more instructors.

Each department must have exactly one instructor serving as its chair. An instructor is not required to chair a department but may chair at most one department.

Each student (STUDENT) has a unique student ID (SId) and a name (SName). The name is a composite attribute consisting of a first name (FName) and a last name (LName). Every student must belong to at least one and at most two departments. A department may have zero or more students.

Each course (COURSE) has a unique course code (CCode) and a unique course name (CoName). Every course must be offered by exactly one department. A department may offer zero or more courses.

A course may have zero or more sections (SECTION), allowing a course to remain in the database even when it has no sections. Each section must belong to exactly one course and must be taught by exactly one instructor. An instructor may teach zero or more sections.

A student may enroll in zero or more sections, and each section must have at least ten students enrolled. Enrollment is represented by the many-to-many relationship TAKES between STUDENT and SECTION.

The database records each student's grade (Grade) for a particular section. Because the grade belongs to a specific student–section combination, Grade is an attribute of TAKES rather than of STUDENT or SECTION alone. This allows a student to receive different grades in different sections.

ER diagram

Please answer on the answer sheet

Part B EER(Enhanced Entity-Relationship) Diagram Matching Exercise

Match each diagram with all applicable statements labeled A–O. A statement may match more than one diagram, and some statements may not be used. There are 15 correct matches in total across the six diagrams.

In all diagrams, a browser cannot also be a buyer or seller. Treat each diagram as an independent model.

A. Every user must have at least one of the three roles. A user may be both a buyer and a seller, while a browser cannot have either of those roles.

B. Penny originally registered on the platform only to browse products and was classified as a BROWSER. After becoming a seller and listing products for sale, Penny can still keep the BROWSER role and hold both roles at the same time.

C. Every user must have exactly one of three roles: buyer, browser, or seller. A user cannot hold more than one role at the same time, and no user is allowed to remain outside all three categories.

D. Anyone who is both a buyer and a seller automatically becomes a platform ambassador. No separate application, review, or appointment is required, because having both roles is enough to qualify.

E. Every platform ambassador must be both a buyer and a seller. Giving up either role means the person can no longer remain an ambassador.

F. Every user must belong to at least one of these categories: BUYER, BROWSER, or SELLER.

G. In a diagram marked d, Ben is allowed to be both a buyer and a seller.

H. A user may be both a buyer and a seller. The model also allows a registered user to remain outside all three roles: buyer, browser, and seller.

I. The double line below USER means that every user must be both a buyer and a seller.

J. In this diagram, Jason can qualify for appointment as a platform ambassador even if he is only a buyer and not a seller.

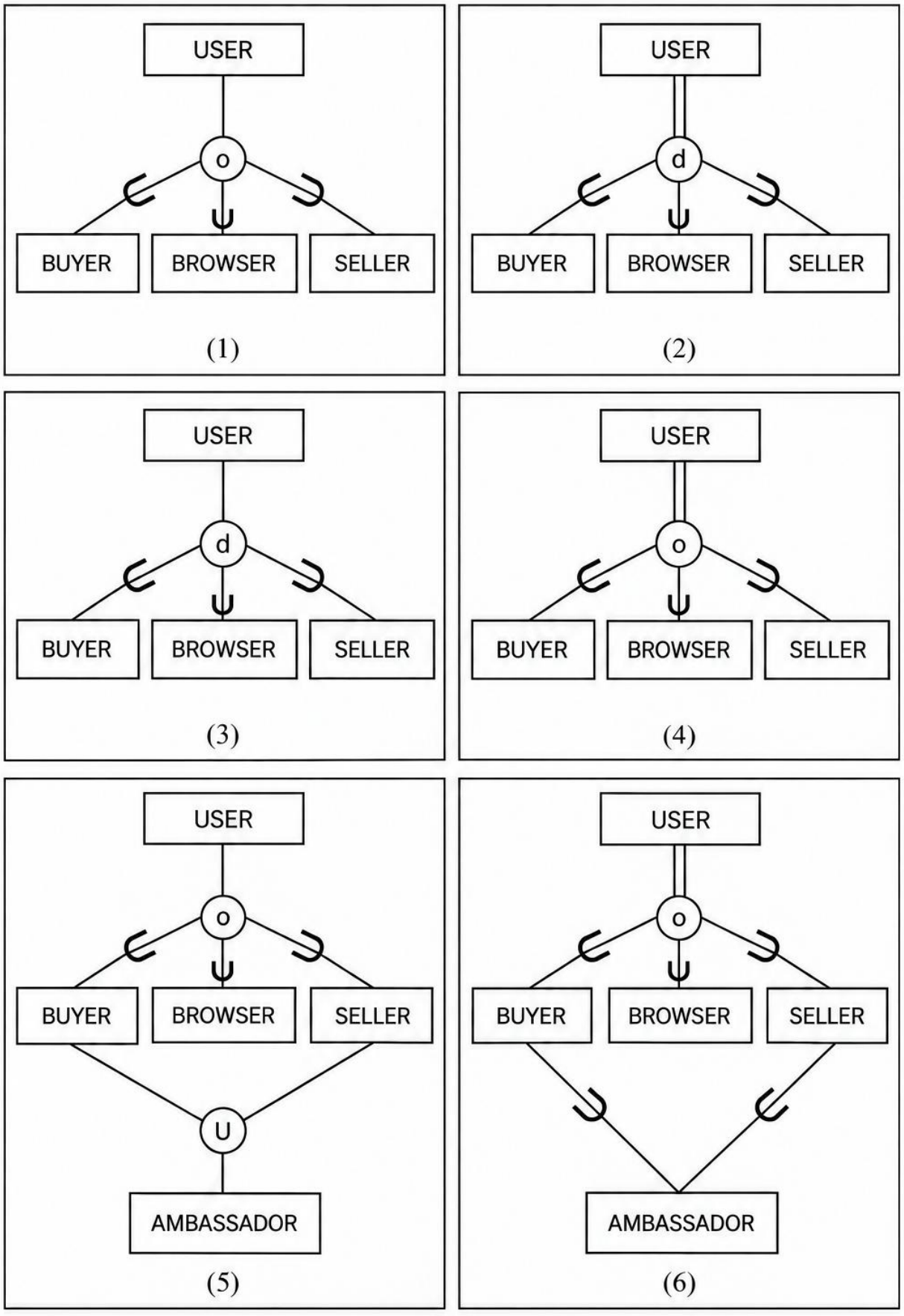
K. The model does not allow any user to be both a buyer and a seller.

L. Pingping is only a browser, but the platform can appoint Pingping as an ambassador without adding a buyer or seller role.

M. A user may have only one of the three roles or none at all, but cannot have more than one role.

N. Every registered user must have all three roles: buyer, browser, and seller.

O. The diagram includes platform ambassadors, but someone who is only a browser cannot become an ambassador.



Matching Exercise

Please answer on the answer sheet,clearly indicating the question number for each answer.

Ex. (1) A B (2) C …. (6)A N

Part C ER/EER Diagram Drawing

Please draw the ER/EER Diagram.

A research institution plans to develop a database to manage chemical materials, experiments, researchers, instruments, and material storage locations. Based on the following requirements, draw an ER/EER diagram showing identifying attributes, relationship cardinalities, and specialization constraints.

Each chemical material (Material) is tracked as an individual container or sample unit and has a unique material identifier (MID) and a material name (MName). A material may be a reagent (Reagent), a sample to be tested (Sample), or both. For example, the same material may be used as a reagent in an experiment and also undergo quality testing. A material whose purpose has not yet been assigned may belong to neither category. Each reagent has a recorded purity (Purity), and each sample has an additional unique sample identifier (SID).

Each laboratory (Lab) has a unique laboratory identifier (LabID). Each researcher (Researcher) has a unique researcher identifier (RID) and must belong to exactly one laboratory. A laboratory may have zero or more researchers.

Each experiment (Experiment) has a unique experiment identifier (EID) and an experiment date (EDate). Each experiment must be managed by exactly one laboratory, and a laboratory may manage zero or more experiments. Each experiment must also be conducted by exactly one researcher, and a researcher may conduct zero or more experiments.

Every experiment managed by this system must use at least one reagent and may use multiple reagents. A reagent may be used in zero or more experiments. Each experiment must test exactly one sample, while a sample may be tested in zero or more experiments.

Each instrument (Instrument) has a unique instrument identifier (InsID) and must belong to exactly one laboratory. A laboratory may own zero or more instruments. Each experiment must use at least one instrument and may use multiple instruments. An instrument may be used in zero or more experiments.

Each storage cabinet (Cabinet) has a unique cabinet identifier (CID) and must be managed by exactly one laboratory. A laboratory may manage zero or more storage cabinets. The database records only the current storage location of each material. A material may have no assigned storage cabinet or may be stored in exactly one cabinet. A storage cabinet may contain zero or more materials.

Hint : store、test、use、manage、own、conduct、belong、organize

Instrument(儀器)、reagent(試劑)、cabinet(櫥櫃)、purity(純淨)

ER/EER diagram

Please answer on the answer sheet